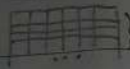


Assignment

data set: มาแล้วครับ



ขั้นตอนที่ 1 คำนวณค่า entropy attribute ที่เลือก Class เลือก attribute ที่เลือก

Age

Credit_card: yes $\frac{3}{7} \approx 0.43$

Credit_card: no $\frac{4}{7} \approx 0.57$

20..40 = $\frac{3}{7} \approx 0.43$

>40 = $\frac{2}{7} \approx 0.29$

<=20 = $\frac{2}{7} \approx 0.29$

20..40 - y = $\frac{2}{3} \approx 0.67$

>40 - y = $\frac{1}{2} = 0.5$

<=20 - y = $\frac{0}{2} = 0$

20..40 - n = $\frac{1}{3} \approx 0.33$

>40 - n = $\frac{1}{2} = 0.5$

<=20 - n = $\frac{2}{2} = 1$

$$\text{entropy (Parent)} = -P(y) \times \log_2 P(y) - P(n) \times \log_2 P(n)$$

$$= -[0.43 \times \log_2 (0.43) + 0.57 \times \log_2 (0.57)]$$

$$= -[0.43 \times (-1.22) + 0.57 \times (-0.81)]$$

$$= 0.99$$

$$\text{entropy (Age = 20..40)} = -[0.67 \times \log_2 (0.67) + 0.33 \times \log_2 (0.33)]$$

$$= -[0.67 \times (-0.58) + 0.33 \times (-1.60)]$$

$$= 0.91$$

$$\text{entropy (Age > 40)} = -[0.5 \times \log_2 (0.5) + 0.5 \times \log_2 (0.5)]$$

$$= 1$$

$$\text{entropy (Age <= 20)} = -[0 \times \log_2 (0) + 1 \times \log_2 (1)]$$

$$= -[0 - 1 + 1 - 0] = 0$$

$$IG (\text{Parent, child}) = \text{entropy (Parent)} - [P(20..40) \times \text{entropy (20..40)} + P(>40) \times \text{entropy (>40)}]$$

$$+ P(<=20) \times \text{entropy (<=20)}]$$

$$= 0.99 - [(0.43 \times 0.91) + (0.29 \times 1) + (0.29 \times 0)]$$

$$= 0.99 - 0.68 = 0.31$$

Income

Credit_card: yes $\frac{3}{7} \approx 0.43$

Credit_card: no $\frac{4}{7} \approx 0.57$

High = $\frac{3}{7} \approx 0.43$

Medium = $\frac{2}{7} \approx 0.29$

Low = $\frac{2}{7} \approx 0.29$

H - y = $\frac{2}{3} \approx 0.67$

M - y = $\frac{1}{2} = 0.5$

L - y = $\frac{0}{2} = 0$

H - n = $\frac{1}{3} \approx 0.33$

M - n = $\frac{1}{2} = 0.5$

L - n = $\frac{2}{2} = 1$

$$\text{entropy}(\text{Income} = H) = -[0.67 \cdot \log_2(0.67) + 0.33 \cdot \log_2(0.33)]$$

$$= 0.91$$

$$\text{entropy}(\text{Income} = M) = -[0.5 \cdot \log_2(0.5) + 0.5 \cdot \log_2(0.5)]$$

$$= 1$$

$$\text{entropy}(\text{Income} = L) = -[0 \cdot \log_2(0) + 1 \cdot \log_2(1)]$$

$$= 0$$

$$I G(\text{Parent}, \text{child}) = 0.99 - [(0.43 \times 0.91) + (0.29 \times 1) + (0.29 \times 0)]$$

$$= 0.31$$

Status Credit_card = yes Credit_card = no

$$\text{Single} = \frac{4}{7} = 0.57 \quad \text{marry} = \frac{3}{7} = 0.43$$

$$S - y = \frac{2}{4} = 0.5 \quad M - y = \frac{1}{3} = 0.33$$

$$S - n = \frac{2}{4} = 0.5 \quad M - n = \frac{2}{3} = 0.67$$

$$\text{entropy}(\text{Status} = S) = -[0.5 \cdot \log_2(0.5) + 0.5 \cdot \log_2(0.5)]$$

$$= 1$$

$$\text{entropy}(\text{Status} = M) = -[0.33 \cdot \log_2(0.33) + 0.67 \cdot \log_2(0.67)]$$

$$= 0.91$$

$$I G(\text{Parent}, \text{child}) = 0.99 - [(0.57 \times 1) + (0.43 \times 0.91)]$$

$$= 0.03$$

Education Credit_card = yes Credit_card = no

$$\text{Master} = \frac{3}{7} = 0.43 \quad \text{Bachelor} = \frac{4}{7} = 0.57$$

$$M - y = \frac{3}{3} = 1 \quad B - y = \frac{0}{4} = 0$$

$$M - n = \frac{0}{3} = 0 \quad B - n = \frac{4}{4} = 1$$

$$\text{entropy}(\text{Education} = M) = -[1 \cdot \log_2(1) + 0 \cdot \log_2(0)]$$

$$= 0$$

$$\text{entropy}(\text{Education} = B) = -[0 \cdot \log_2(0) + 1 \cdot \log_2(1)]$$

$$= 0$$

$$I G(\text{Parent}, \text{child}) = 0.99 - [(0.43 \times 0) + (0.57 \times 0)]$$

$$= 0.99$$

Summarize Results

IG - Age = 0.31

IG - Income = 0.31

IG - Status = 0.03

IG - Education = 0.99 \ \ Root node

Decision Rules

if Education : Master then Credit_card : Yes

if Education : Bachelor then Credit_card : No #